Existing App Survey & Comparison

VNU-HCM Library App vs. CERN Library Catalogue

Software Engineering Project - PA1

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1 Introduction

As part of the project proposal phase, this report surveys two existing library management applications: the VNU-HCM Library Mobile App and the CERN Library Catalogue. The objective is to understand existing solutions, evaluate their user interface and workflow design, and identify opportunities for differentiation for our proposed application.

2 Common Features

Based on the survey of both the VNU-HCM Library App and the CERN Library Catalogue, several core functional similarities were identified:

- **Core Search Functionality:** Both systems provide a prominent search bar as the primary tool for discovering books, documents, and journals.
- **Detailed Physical Copy Tracking:** Both applications display comprehensive details about physical copies, including title, author, shelving location (e.g., "CL-General Collection" or specific call numbers), barcode, and current availability status (e.g., "Available" or "On shelf").
- **Direct Borrowing Actions:** Users can initiate a loan request directly from the item details page using primary action buttons such as "Borrow" (VNU) or "Request loan" (CERN).

3 Proposed Differentiators (What Our App Will Do Better)

While the existing apps provide solid foundational features, our proposed application will introduce several improvements to address their current limitations:

- **Optimized Screen Space Utilization:** Both existing apps suffer from poor space management—VNU leaves significant empty space below the digital card, while CERN utilizes an overly large hero background image. Our app will replace these empty or purely decorative areas with highly functional content, such as active loan summaries, recent activity history, or personalized recommendations.

- **Refined and Accessible Search Filters:** The VNU app lacks advanced filtering options on the immediate search results screen, whereas CERN provides a filter system that is extremely dense and potentially overwhelming for new users. Our solution will implement a simplified, collapsible filter panel that provides essential refinements (e.g., by campus, year, or availability) without cluttering the mobile interface.

- **Actionable Empty States and Error Handling:** The VNU app handles missing data poorly by simply displaying a blank "No data" screen. Our app will feature actionable empty states, providing users with clear contextual feedback, suggestions to reload, or steps to refine their search inputs to prevent workflow disruption.

4 Adopted UI/UX Patterns

To ensure a highly intuitive and efficient user experience, our application will adopt the following successful UI/UX patterns from the surveyed apps:

- **Search-First Design (from CERN):** We will place the search bar prominently on the home screen to make the primary task of information discovery immediately accessible.

- **Card-Based UI (from VNU):** We will use distinct visual content blocks (cards) for search results and item details, which optimizes the display of complex bibliographic information within a confined mobile screen space.

- **Persistent Navigation (from VNU):** A bottom navigation bar will be implemented to allow users to quickly switch between core features (Home, Search, Profile) without getting lost in deep menus.

- **Primary Action Emphasis (from both):** Crucial actions like the "Borrow" button will be visually distinct, spanning a large area, and anchored consistently (e.g., at the bottom of the screen) to guide the user seamlessly through the borrowing task.